

**Determinants and Forecasting of UK
Inflation Pre and Post Covid19 Pandemic:
An Econometric Analysis (2015-2025)**

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1. Introduction

The rate of inflation is one of the most significant indicators of macroeconomic stability, influencing consumer purchasing power, business investment, monetary policy, and overall economic growth. In the United Kingdom, inflation has become an increasingly important area of economic concern following the COVID-19 pandemic and subsequent global supply-side disruptions. Between 2021 and 2023, the UK experienced some of its highest inflation rates in decades, driven by a combination of rising energy prices, labour market pressures, supply chain disruptions, and shifts in consumer demand. As inflation accelerated well above the Bank of England's target rate, understanding both the determinants and future trajectory of inflation became increasingly important for policymakers, businesses, and households.

Economic theory suggests that inflation can be influenced by several macroeconomic factors. Traditional Phillips Curve theory proposes an inverse relationship between unemployment and inflation, while cost-push theories emphasize the role of production costs and external shocks such as energy price increases. In addition, wage growth and consumer expectations may contribute to inflation persistence through demand-side and behavioral mechanisms. However, the post-pandemic inflationary environment has challenged many conventional economic assumptions, creating renewed interest in empirical analysis of inflation dynamics and forecasting performance.

This project investigates the key determinants of UK inflation between 2015 and 2025 using a range of econometric techniques. Monthly data obtained from the Office for National Statistics (ONS) and related macroeconomic sources are used to examine the relationship between Consumer Price Index (CPI) inflation, wage growth, unemployment, energy prices, and consumer confidence. Multiple linear regression models are employed to estimate the significance and direction of these relationships, while diagnostic testing, robust standard errors, stationarity testing, and dynamic modelling are used to improve the reliability of the analysis.

In addition to explanatory econometric modeling, this project also explores inflation forecasting using time-series techniques. An ARIMA forecasting model is developed to analyse inflation persistence and evaluate the model's ability to predict post-2023 inflation trends. Forecast performance is assessed using out-of-sample testing and forecast accuracy measures, allowing comparison between predicted and observed inflation rates during the post-pandemic period.

The findings of this project contribute to a broader understanding of UK inflation dynamics during a period of substantial economic uncertainty and structural change. Recent geopolitical tensions in the Middle East, including concerns surrounding Iranian oil supply disruptions and instability in the Strait of Hormuz, have further reinforced the vulnerability of inflation dynamics to external energy shocks. Such developments create significant challenges for central banks, as policymakers must balance price stability against the preservation of economic growth amidst increasing pressure of stagflation. Furthermore, the analysis highlights both the usefulness and limitations of econometric forecasting models when applied to volatile macroeconomic environments characterised by external shocks, structural breaks, and rapidly changing economic conditions.

2. Literature Review

The relationship between inflation, labour market conditions, and macroeconomic stability has long occupied a central position within economic theory and monetary policy analysis. One of the most influential frameworks used to explain inflation dynamics is the Phillips Curve, originally proposed by A.W. Phillips (1958), which suggested an inverse relationship between unemployment and wage inflation. Over time, this relationship evolved into the modern New Keynesian Phillips Curve (NKPC), which incorporates inflation expectations, forward-looking behaviour, and supply-side factors into inflation determination. Despite extensive empirical debate surrounding the stability of the Phillips Curve, it remains highly influential in both macroeconomic theory and practical monetary policymaking.

Batini, Jackson and Nickell (2005) extend the traditional Phillips Curve framework by developing an open-economy New Keynesian Phillips Curve model for the United Kingdom. Their analysis demonstrates that UK inflation is influenced not only by labour market conditions and employment dynamics, but also by external cost pressures such as import prices and oil prices. In particular, the authors find that increases in labour's share and real oil prices contribute positively to inflationary pressures, highlighting the importance of both domestic and external supply-side determinants of inflation. These findings are particularly relevant within the context of the post-pandemic inflation surge, during which global energy price shocks and labour market disruptions contributed significantly to rising UK inflation. Consequently, the inclusion of wage growth, unemployment, and energy prices within this project's econometric framework is strongly supported by both macroeconomic theory and empirical evidence.

The persistence and evolving behaviour of inflation have also become increasingly important within macroeconomic research. Osborn and Sensier (2009) argue that the statistical properties of UK inflation have changed substantially over time, particularly following the introduction of inflation targeting in the early 1990s. Their findings suggest that inflation persistence, seasonality, and forecasting performance are highly sensitive to changes in monetary policy regimes and broader macroeconomic conditions. Importantly, the authors conclude that inflation should not be treated as a constant parameter process, as structural breaks and regime changes can significantly alter inflation dynamics over time. This perspective is especially relevant in the aftermath of the COVID-19 pandemic, during which inflation behaviour became considerably more volatile and difficult to predict. The presence of structural breaks within inflation data provides strong justification for the inclusion of post-COVID interaction models, stationarity testing, and dynamic inflation modelling within this project.

More recent literature has focused specifically on the inflationary dynamics observed following the COVID-19 pandemic and the associated global supply-side disruptions. Harding, Lindé and Trabandt (2023) propose a nonlinear Phillips Curve model capable of explaining both the relatively modest inflation response during the Global Financial Crisis and the sharp inflation surge observed during the post-COVID period. Their analysis suggests that inflationary pressures become amplified when inflation is already elevated, leading to stronger transmission of cost-push shocks and increased inflation volatility. Importantly, the authors argue that central banks face more severe trade-offs between inflation stabilisation and output stabilisation during periods of elevated inflation. This insight is particularly relevant given the substantial increase in global energy prices, supply chain disruptions, and geopolitical instability observed after 2020.

The role of energy prices and external shocks has become increasingly important within modern inflation analysis. Supply-side disruptions associated with the COVID-19 pandemic, the Russia-Ukraine conflict, and

more recent geopolitical tensions in the Middle East have reinforced the vulnerability of inflation dynamics to fluctuations in global commodity markets. Rising oil and energy prices directly increase production and transportation costs across the economy, contributing to cost-push inflationary pressures. Furthermore, instability surrounding key energy transit routes such as the Strait of Hormuz has increased concerns regarding future inflation volatility and energy security. These developments highlight the extent to which modern inflation dynamics are increasingly shaped by external and geopolitical factors rather than purely domestic demand conditions.

The implications of inflation persistence and supply-side inflation shocks also present significant challenges for monetary policymakers. Sentance (2007) argues that successful monetary policy requires maintaining well-anchored inflation expectations while balancing demand conditions against the productive capacity of the economy. However, when inflation is driven by external supply-side shocks rather than excessive aggregate demand, central banks may face substantial policy trade-offs. Raising interest rates may reduce inflationary pressures over time but can simultaneously weaken economic growth and labour market performance. Consequently, modern monetary policy increasingly involves balancing inflation stability against broader macroeconomic objectives amidst heightened uncertainty and stagflationary risks.

Overall, the existing literature suggests that inflation dynamics are shaped by a complex interaction of labour market conditions, inflation expectations, energy prices, external supply shocks, and monetary policy regimes. Moreover, the post-COVID inflationary environment has demonstrated that inflation processes are neither stable nor constant over time, creating additional challenges for econometric forecasting and policy analysis. Building upon this literature, the present study empirically investigates the determinants of UK inflation between 2015 and 2025 using regression analysis, dynamic modelling, and ARIMA forecasting techniques to evaluate both inflation drivers and forecasting performance during a period of substantial economic disruption and structural change.

3. Data and Methodology

3.1 Data

This study employs monthly UK macroeconomic data covering the period from January 2015 to December 2025 to analyse the determinants and forecasting performance of UK inflation during both stable and highly volatile economic conditions. The selected sample period captures several major macroeconomic events, including the post-Brexit adjustment period, the COVID-19 pandemic, the global energy price shock, and the subsequent inflation surge observed across advanced economies. The use of monthly data allows for a more detailed analysis of short-run inflation dynamics and improves the reliability of time-series forecasting techniques.

The dependent variable used throughout the analysis is the UK Consumer Price Index (CPI) annual inflation rate, which serves as the primary measure of inflation within the United Kingdom. To examine the determinants of inflation, four explanatory variables are included within the econometric framework: wage growth, unemployment, energy prices, and consumer confidence. These variables were selected based on their relevance within both macroeconomic theory and the existing empirical literature surrounding inflation dynamics.

Data for CPI inflation, wage growth, and unemployment were obtained from the Office for National Statistics (ONS), which provides official UK macroeconomic and labour market statistics. Energy price data were proxied using Brent crude oil prices, reflecting the importance of global energy costs within modern inflationary dynamics. Consumer confidence data were included to capture changes in household sentiment and demand-side conditions within the economy. All datasets were cleaned, standardised, and merged into a single monthly dataset using RStudio in preparation for econometric analysis and forecasting procedures.

The inclusion of these variables reflects the study’s objective of capturing both demand-side and supply-side determinants of inflation. Wage growth and unemployment are incorporated to reflect labour market conditions and Phillips Curve dynamics, while energy prices capture the role of external cost-push inflationary pressures. Consumer confidence is included as an indicator of household expectations and aggregate demand conditions. Together, these variables provide a broad macroeconomic framework through which UK inflation behaviour can be examined over the sample period.

Table 1: Variables and their sources

VARIABLE	DESCRIPTION	SOURCE
CPI	UK Consumer Price Index annual inflation rate (%)	Office for National Statistics (ONS)
WAGE GROWTH	UK average wage growth (%)	Office for National Statistics (ONS)
UNEMPLOYMENT	UK unemployment rate (%)	Office for National Statistics (ONS)

ENERGY PRICES	Brent crude oil prices (USD per barrel)	Investing.com / FRED
CONSUMER CONFIDENCE	UK consumer confidence index	GFK / OECD

The selected variables were chosen to reflect both traditional and modern explanations of inflation dynamics within the United Kingdom. Wage growth and unemployment were included to capture labour market conditions and Phillips Curve relationships; whereby tighter labour markets may contribute to upward wage and inflationary pressures. Energy prices were incorporated to account for external supply-side shocks and cost-push inflation, particularly given the significant role played by global oil and energy markets during the post-pandemic inflation surge. Consumer confidence was included as a proxy for household sentiment and aggregate demand conditions, as stronger consumer confidence may contribute to increased spending and inflationary pressures within the economy. Collectively, these variables allow for the analysis of both domestic macroeconomic conditions and external inflationary shocks across the sample period. Although interest rates represent an important component of monetary policy transmission and inflation control, they were not included within the final econometric specification. This decision was primarily due to limitations surrounding data consistency and comparability across the selected sample period, particularly when integrating monthly macroeconomic variables from multiple sources. Furthermore, the study focuses primarily on observable labour market conditions, supply-side cost pressures, and consumer sentiment as direct determinants of inflation dynamics. Nevertheless, the influence of monetary policy remains highly relevant within the broader discussion of inflation persistence and central bank policy trade-offs explored throughout the analysis.

Table 2: Correlation Matrix of Inflation and Explanatory Variables

	CPI	Wage Growth	Unemployment	Energy Prices	Consumer Confidence
CPI	1.00	0.65	-0.59	0.75	-0.79
Wage Growth	0.65	1.00	-0.33	0.74	-0.40
Unemployment	-0.59	-0.33	1.00	-0.59	0.47
Energy Prices	0.75	0.74	-0.59	1.00	-0.52
Consumer Confidence	-0.79	-0.40	0.47	-0.52	1.00

Source: Author's calculations using ONS, Brent Oil Price, and OECD data

3.2 Econometric Framework

To examine the determinants of UK inflation, this study employs a multiple linear regression framework estimated using Ordinary Least Squares (OLS) techniques within RStudio. The primary objective of the regression analysis is to estimate the relationship between CPI inflation and a range of macroeconomic explanatory variables, including wage growth, unemployment, energy prices, and consumer confidence. Multiple regression analysis allows the simultaneous estimation of the individual effect of each explanatory variable on inflation while holding other variables constant.

The baseline econometric specification used throughout the analysis is given by:

$$CPI_t = \beta_0 + \beta_1 WageGrowth_t + \beta_2 Unemployment_t + \beta_3 EnergyPrices_t + \beta_4 ConsumerConfidence_t + \varepsilon_t$$

Where our dependent variable is given by CPI_t which represents the UK inflation rate at the time given t , β_0 denotes the intercept term, and β_1 to β_4 represent the estimated coefficients associated with each explanatory variable. The error term ε_t captures the unexplained variation in inflation not accounted for by the model.

Within this framework, the estimated coefficients measure the expected change in CPI inflation associated with a one-unit increase in an explanatory variable, holding all other variables constant. For example, a positive coefficient on energy prices would suggest that increases in oil prices are associated with higher inflation rates, consistent with theories of cost-push inflation. Similarly, unemployment and wage growth variables allow for the examination of Phillips Curve dynamics and labour market inflationary pressures within the UK economy.

In addition to the baseline regression model, this study also incorporates a post-COVID dummy variable to examine whether inflation dynamics changed following the onset of the COVID-19 pandemic in March 2020. The dummy variable takes the value of 0 for observations prior to March 2020 and 1 for observations thereafter. Interaction terms between the dummy variable and the explanatory variables were included to investigate whether the relationships between inflation and its determinants strengthened or weakened during the post-pandemic inflationary environment.

Several diagnostic tests were conducted to assess the reliability and validity of the regression results. Variance Inflation Factor (VIF) tests were used to examine potential multicollinearity between explanatory variables, while heteroskedasticity-consistent robust standard errors were employed to improve the reliability of statistical inference. Autocorrelation within the residuals was examined using the Durbin-Watson test, and Augmented Dickey-Fuller (ADF) tests were conducted to assess the stationarity properties of the inflation series. Where appropriate, differencing procedures and dynamic modelling techniques were implemented to address persistence and non-stationarity within the data.

Following the regression analysis, additional time-series forecasting techniques were employed to evaluate inflation persistence and forecasting performance over the sample period. Specifically, an Autoregressive Integrated Moving Average (ARIMA) model was estimated using historical CPI inflation data. The dataset was divided into training and testing samples, with observations from 2015 to 2023 used to estimate the forecasting

model and 2024 data reserved for out-of-sample forecast evaluation. Forecast accuracy was subsequently assessed by comparing predicted inflation values against realised inflation outcomes.

4. Empirical Results and Analysis

4.1 Baseline Regression results

Table 3: Baseline OLS Regression Results for UK CPI Inflation

Variable	Coefficient	Std.Error	t-Statistic	p-value
Intercept	1.369	1.800	0.760	0.449
Wage Growth	0.339***	0.090	3.750	0.000
Unemployment	-0.828**	0.323	-2.567	0.012
Energy Prices	0.033***	0.012	2.811	0.006
Consumer Confidence	-0.149***	0.014	-10.361	0.000

Model Statistics

Statistic	Value
Observations	119
R ²	0.806
Adjusted R ²	0.799
F-Statistic	118.5
Prob > F	<0.001

Notes: * $p < 0.01$, ** $p < 0.05$, * $p < 0.10$**

Source: Author's calculations using RStudio.

Table 3 presents the baseline Ordinary Least Squares (OLS) regression results examining the determinants of UK CPI inflation between 2015 and 2025. Overall, the model demonstrates strong explanatory power, with an R^2 value of 0.806 indicating that approximately 80.6% of the variation in UK inflation is explained by the explanatory variables included within the regression. The adjusted R^2 value of 0.799 further suggests that the model maintains strong explanatory performance after accounting for the number of explanatory variables included within the specification.

In addition, the overall regression model is highly statistically significant, as indicated by the F-statistic of 118.5 and the associated probability value below 0.001. This suggests that the explanatory variables jointly provide substantial explanatory power in accounting for movements in UK inflation over the sample period.

The estimated coefficients presented within Table 3 provide several important insights into the determinants of UK inflation over the sample period. Wage growth exhibits a positive and statistically significant relationship with CPI inflation, suggesting that stronger wage increases contribute to higher inflationary pressures within the economy. Specifically, a one-unit increase in wage growth is associated with an approximate 0.339 percentage point increase in CPI inflation, holding all other variables constant. This finding is broadly consistent with traditional Phillips Curve theory and the wider literature surrounding wage-price dynamics, whereby tighter labour market conditions and rising nominal wages contribute to increased inflationary pressures through higher production costs and stronger aggregate demand.

Unemployment displays a negative and statistically significant coefficient, consistent with Phillips Curve relationships observed throughout macroeconomic literature. The estimated coefficient implies that higher unemployment rates are associated with lower inflation outcomes, reflecting weaker labour market conditions and reduced demand-side pressures within the economy. This finding aligns closely with the work of Batini, Jackson and Nickell (2005), who identify labour market conditions as an important determinant of UK inflation dynamics within an open-economy New Keynesian Phillips Curve framework.

Energy prices also demonstrate a positive and statistically significant relationship with inflation, supporting theories of cost-push inflation and external supply-side pressures. The positive coefficient indicates that increases in Brent crude oil prices are associated with higher UK inflation rates, highlighting the extent to which global commodity markets and external energy shocks influence domestic inflation dynamics. This result is particularly relevant within the post-pandemic inflationary environment, during which rising global energy prices and geopolitical instability contributed substantially to inflation volatility across advanced economies. The findings therefore reinforce arguments presented within both Batini et al. (2005) and Harding, Lindé and Trabandt (2023), who emphasise the importance of external cost-push shocks and energy prices in shaping modern inflation dynamics.

Consumer confidence exhibits a strong negative and highly statistically significant relationship with CPI inflation. This result may initially appear counterintuitive, as stronger consumer sentiment is often associated with increased aggregate demand and inflationary pressures. However, the negative coefficient likely reflects the sharp deterioration in household confidence observed during periods of elevated inflation and economic

uncertainty following the COVID-19 pandemic and energy price shock. In this context, rising inflation may have eroded real household purchasing power and weakened consumer expectations regarding future economic conditions. Consequently, the results suggest that periods of heightened inflation volatility were associated with declining consumer confidence across the UK economy.

Overall, the baseline regression results provide substantial empirical support for both traditional Phillips Curve relationships and modern supply-side explanations of inflation. The findings suggest that UK inflation dynamics between 2015 and 2025 were influenced by a combination of labour market conditions, external energy shocks, and broader macroeconomic sentiment. Furthermore, the strong explanatory power of the regression model indicates that the selected variables collectively capture a substantial proportion of variation in UK inflation over the sample period.

4.2 Diagnostic Testing and Model Validation

Table 4: Variance Inflation Factor (VIF) Results

Variable	VIF
Wage Growth	2.351
Unemployment	1.713
Energy Prices	3.359
Consumer Confidence	1.477

Source: Author’s calculations using RStudio.

Table 5: Diagnostic Test Results

Test	Statistic	p-value	Interpretation
Durbin-Watson	0.906	<0.001	Positive autocorrelation present
ADF Test (Level CPI)	-1.799	0.660	Non – stationary

ADF Test (Differenced CPI)	-4.963	0.010	Stationary after differencing
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Source: Author’s calculations using RStudio.

Following the estimation of the baseline regression model, several diagnostic tests were conducted to evaluate the reliability and robustness of the econometric results. Given the time-series nature of the data, it was important to assess potential issues relating to multicollinearity, autocorrelation, and non-stationarity, all of which may affect the validity of Ordinary Least Squares (OLS) estimation and statistical inference.

Table 4 presents the Variance Inflation Factor (VIF) results for the explanatory variables included within the baseline regression model. The reported VIF values remain relatively low and comfortably below conventional critical thresholds commonly associated with severe multicollinearity. This suggests that the explanatory variables do not exhibit excessive correlation with one another and that the estimated coefficients remain sufficiently reliable for interpretation. Although wage growth and energy prices demonstrate moderate correlation, the VIF results indicate that multicollinearity is not sufficiently severe to undermine the stability of the regression estimates.

Table 5 reports the results of the Durbin-Watson and Augmented Dickey-Fuller (ADF) diagnostic tests. The Durbin-Watson statistic of 0.906 indicates the presence of positive autocorrelation within the regression residuals, suggesting that inflation dynamics exhibit substantial persistence over time. This finding is consistent with macroeconomic theory and previous literature examining inflation persistence within the United Kingdom, particularly during periods characterised by prolonged supply-side shocks and elevated inflation expectations.

The Augmented Dickey-Fuller test results further indicate that the CPI inflation series is non-stationary in levels, as the null hypothesis of a unit root could not initially be rejected. Non-stationary time series may generate spurious regression relationships and unreliable statistical inference if left unaddressed. Consequently, first differencing procedures were implemented to stabilise the mean of the series and remove persistence effects. Following differencing, the ADF test strongly rejected the null hypothesis of non-stationarity, indicating that the transformed inflation series became stationary and therefore more suitable for dynamic time-series modelling.

Collectively, the diagnostic testing procedures suggest that while the baseline regression model provides substantial explanatory power, inflation dynamics within the United Kingdom also exhibit important time-series properties including persistence and autocorrelation. These findings provide additional justification for the subsequent use of dynamic inflation models and ARIMA forecasting techniques to better capture evolving inflation behaviour over the sample period.

4.3 Dynamic and Post-COVID Inflation Analysis

Table 6: Post-COVID Regression Results for UK CPI Inflation

Variable	Coefficient	Std.Error	t-Statistic	p-value
Intercept	8.209**	3.985	2.060	0.045
Wage Growth	0.709***	0.148	4.790	0.000
Unemployment	-2.425***	0.704	-3.442	0.001
Energy Prices	-0.003	0.018	-0.185	0.854
Consumer Confidence	-0.179***	0.020	-8.788	0.000

Model Statistics

Statistic	Value
Observations	56
R ²	0.838
Adjusted R ²	0.825
F-Statistic	66.01
Prob > F	<0.001

Notes: *** p < 0.01, ** p < 0.05, * p < 0.10

Source: Author’s calculations using RStudio.

Table 6 presents the post-COVID regression results examining the determinants of UK CPI inflation following the onset of the COVID-19 pandemic and subsequent inflation surge. The results indicate that inflation dynamics changed substantially during the post-pandemic period, with several explanatory variables exhibiting stronger relationships with inflation relative to the baseline regression model. The model maintains strong explanatory power, with an R^2 value of 0.838 suggesting that approximately 83.8% of the variation in post-COVID UK inflation is explained by the selected explanatory variables.

One of the most notable findings relates to wage growth, which displays a substantially larger positive coefficient within the post-COVID model compared to the baseline regression results. The estimated coefficient implies that a one-unit increase in wage growth is associated with an approximate 0.709 percentage point increase in CPI inflation, holding other variables constant. This suggests that wage-price pressures intensified considerably during the post-pandemic period, likely reflecting labour shortages, supply chain disruptions, and tight labour market conditions experienced across the UK economy following the reopening period after COVID-19 restrictions. The results therefore provide evidence that inflation became increasingly sensitive to domestic labour market pressures during the post-pandemic recovery.

Similarly, unemployment demonstrates a substantially stronger negative relationship with inflation compared to the baseline specification. The larger negative coefficient suggests that labour market tightness exerted a more pronounced influence on inflation dynamics during the post-COVID period. This finding is broadly consistent with recent literature surrounding nonlinear and steepening Phillips Curve relationships, whereby inflation becomes increasingly responsive to labour market conditions during periods of elevated inflationary pressure. In this context, the results align closely with Harding, Lindé and Trabandt (2023), who argue that inflation-output trade-offs become more severe when inflationary pressures are elevated and supply-side shocks propagate more strongly throughout the economy.

In contrast, energy prices lose statistical significance within the post-COVID regression model despite their significance within the baseline specification. One possible explanation is that the direct impact of global energy shocks may already have been absorbed into broader inflation expectations and wage-setting behaviour during the post-pandemic inflation surge. Alternatively, the result may reflect the increasing importance of domestic inflation persistence and labour market dynamics relative to purely external supply-side cost pressures during the later stages of the inflation cycle.

Consumer confidence continues to exhibit a strong negative and highly statistically significant relationship with inflation throughout the post-COVID period. The persistence of this relationship highlights the extent to which elevated inflation and economic uncertainty negatively affected household sentiment and consumer expectations. Rapid increases in living costs, energy bills, and interest rate expectations likely contributed to weaker household confidence during this period, reinforcing the broader macroeconomic instability observed across the UK economy.

Overall, the post-COVID regression results provide strong evidence that UK inflation dynamics became increasingly driven by labour market conditions and inflation persistence following the pandemic period. The findings suggest that inflationary pressures during the post-2020 environment reflected not only external energy shocks, but also broader structural shifts within labour markets, expectations formation, and domestic macroeconomic behaviour. These developments created substantial policy challenges for central banks, which faced increasingly difficult trade-offs between controlling inflation and maintaining economic growth during a period characterised by heightened global uncertainty and supply-side disruption.

Table 7: Interaction Model of Post-COVID Inflation Dynamics

Variable	Coefficient	Std.Error	t-Statistic	p-value
Intercept	7.355***	1.958	3.756	0.000
Wage Growth	-0.153	0.178	-0.862	0.391
Unemployment	-1.407***	0.309	-4.555	0.000
Energy Prices	0.019	0.014	1.393	0.166
Consumer Confidence	0.031	0.034	0.894	0.373
Wage Growth x Post-COVID	0.861***	0.211	4.084	0.000
Unemployment x Post-COVID	-0.870***	0.163	-5.336	0.000
Energy Prices x Post-COVID	-0.020	0.016	-1.289	0.200
Consumer Confidence x Post-COVID	-0.211***	0.036	-5.924	0.000

Model Statistics

Statistics	Value
Observations	119
R ²	0.874
Adjusted R ²	0.865
F-Statistic	95.69
Prob > F	<0.001

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Source: Author's calculations using RStudio.

Table 7 presents the interaction model examining whether the relationships between inflation and its key macroeconomic determinants changed following the onset of the COVID-19 pandemic. The inclusion of interaction terms allows the analysis to capture structural shifts in inflation dynamics during the post-pandemic period, providing a framework conceptually similar to Difference-in-Differences approaches by testing whether the responsiveness of inflation to key explanatory variables intensified or weakened after 2020. Overall, the model demonstrates very strong explanatory performance, with an R^2 value of 0.874 indicating that approximately 87.4% of the variation in UK inflation is explained by the model specification.

The interaction terms reveal substantial evidence that inflation dynamics changed significantly during the post-COVID period. Most notably, the interaction between wage growth and the post-COVID dummy variable is positive and highly statistically significant. This finding suggests that wage growth became substantially more inflationary following the pandemic period, supporting the argument that labour market shortages, recruitment difficulties, and tight employment conditions intensified domestic inflationary pressures across the UK economy. The magnitude of the coefficient implies that wage-price dynamics strengthened considerably after COVID-19, reinforcing concerns surrounding the emergence of wage-price persistence during the inflation surge observed between 2021 and 2023.

Similarly, the interaction between unemployment and the post-COVID period is negative and highly statistically significant, indicating that inflation became increasingly sensitive to labour market conditions following the pandemic. This result provides strong evidence of a steepening Phillips Curve relationship during periods of elevated inflationary pressure. In practical terms, tighter labour markets and lower unemployment rates exerted a substantially larger upward effect on inflation during the post-pandemic environment relative to the pre-COVID period. These findings align closely with Harding, Lindé and Trabandt (2023), who argue that inflation-output trade-offs become increasingly severe when inflation expectations become less stable and supply-side shocks propagate more strongly throughout the economy.

The interaction term associated with consumer confidence also demonstrates a large negative and statistically significant effect. This result suggests that the relationship between inflation and household sentiment deteriorated considerably during the post-pandemic inflation surge. Elevated inflation, rising living costs, and uncertainty surrounding future economic conditions likely contributed to declining consumer confidence across the UK economy. The findings therefore reinforce the broader argument that post-pandemic inflation dynamics were not solely driven by external commodity prices, but also reflected shifts in expectations formation, household behaviour, and domestic macroeconomic uncertainty.

In contrast, the interaction term associated with energy prices does not remain statistically significant within the interaction specification. While energy prices were highly important during the initial stages of the inflation surge, the findings may indicate that subsequent inflation persistence became increasingly driven by domestic wage-setting behaviour and labour market conditions rather than purely external commodity shocks. This result is particularly important because it suggests that inflationary pressures evolved over time, transitioning from externally driven cost-push inflation toward more persistent and internally embedded inflation dynamics.

Collectively, the interaction model provides some of the strongest evidence within this study regarding the structural transformation of UK inflation dynamics following COVID-19. The results suggest that the post-pandemic inflationary environment was characterised by stronger labour market transmission effects, heightened inflation persistence, and increased macroeconomic sensitivity to domestic conditions. These findings carry important implications for monetary policymakers, as they suggest that inflation control became increasingly difficult once inflation expectations and wage-setting dynamics are adjusted to the high-inflation environment. Consequently, the results highlight the complex trade-offs faced by central banks attempting to stabilise inflation while simultaneously avoiding excessive damage to employment and economic growth during periods of heightened macroeconomic instability.

Table 8: Dynamic Inflation Model

Variable	Coefficient	Std. Error	t-Statistic	p-value
Intercept	1.236	1.654	0.747	0.456
Lagged CPI	0.356***	0.067	5.308	0.000
Wage Growth	0.233**	0.084	2.760	0.007
Unemployment	-0.623**	0.302	-2.061	0.042
Energy Prices	0.020*	0.011	1.851	0.067

Consumer Confidence	-0.101***	0.016	-6.353	0.000
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Model Statistics

Statistic	Value
Observations	118
R ²	0.845
Adjusted R ²	0.838
F-Statistic	122.1
Prob > F	< 0.001

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Source: Author's calculations using RStudio.

Table 8 presents the dynamic inflation model incorporating lagged CPI inflation in order to examine the persistence of inflationary pressures within the UK economy. The inclusion of lagged inflation allows the model to capture the extent to which previous inflation levels influence current inflation outcomes, thereby reflecting the role of inflation expectations, price-setting behaviour, and persistence dynamics over time. The model demonstrates strong explanatory power, with an R² value of 0.845 indicating that approximately 84.5% of the variation in UK inflation is explained by the dynamic specification.

One of the most significant findings within the model is the positive and highly statistically significant coefficient associated with lagged CPI inflation. The estimated coefficient suggests that previous inflation levels exert a substantial influence on current inflation outcomes, providing strong empirical evidence of inflation persistence within the United Kingdom over the sample period. This finding implies that inflationary shocks were not purely temporary or transitory in nature but instead became partially embedded within broader macroeconomic behaviour and expectations formation. Such persistence is particularly important from a monetary policy perspective, as persistent inflation is considerably more difficult for central banks to stabilise once inflation expectations begin adjusting to higher price growth environments.

The evidence of inflation persistence identified within the dynamic model aligns closely with the work of Osborn and Sensier (2009), who argue that UK inflation dynamics exhibit important structural changes and persistence effects following shifts in monetary policy regimes and inflation expectations. The results are also consistent with broader New Keynesian inflation literature, which emphasises the importance of backward -

looking inflation behaviour, adaptive expectations, and delayed price adjustment mechanisms during periods of macroeconomic instability.

The remaining explanatory variables continue to display relationships broadly consistent with the baseline and interaction model specifications. Wage growth retains a positive and statistically significant relationship with inflation, suggesting that labour market pressures remained an important driver of domestic inflationary dynamics. Unemployment continues to exhibit a negative relationship with inflation, further supporting Phillips Curve theory and reinforcing the argument that tighter labour market conditions contributed to inflation persistence during the post-pandemic period.

Energy prices remain positively associated with inflation, although the coefficient becomes comparatively weaker within the dynamic specification. This result may suggest that while external energy shocks initially contributed to the inflation surge, inflation persistence increasingly became driven by domestic behavioural and institutional mechanisms such as wage-setting behaviour, inflation expectations, and delayed price adjustment processes. In this context, external supply-side shocks may have acted as the initial trigger for inflation acceleration, while persistence mechanisms contributed to sustaining elevated inflation over time.

Consumer confidence continues to display a strong negative relationship with inflation, reflecting the deterioration in household sentiment during periods characterised by elevated inflation volatility and economic uncertainty. Rising living costs, declining real incomes, and uncertainty surrounding future monetary policy conditions likely contributed to weaker confidence across UK households throughout the sample period.

Collectively, the dynamic inflation model provides strong evidence that UK inflation between 2015 and 2025 was characterised not only by external supply-side shocks and labour market pressures, but also by substantial inflation persistence effects. These findings carry important implications for policymakers, as persistent inflation environments create increasingly difficult trade-offs between inflation stabilisation and economic growth. In particular, geopolitical tensions surrounding global energy markets, including concerns relating to Middle Eastern oil supply disruptions and instability within key shipping routes such as the Strait of Hormuz, further reinforce the vulnerability of inflation dynamics to prolonged external shocks. Under such conditions, central banks may face heightened stagflationary risks, whereby efforts to reduce inflation through tighter monetary policy simultaneously weaken output growth and labour market conditions. Consequently, the findings highlight the increasingly complex policy environment confronting modern central banks during periods of persistent inflation and heightened global uncertainty.

4.4 ARIMA Forecasting Analysis

in addition to the regression-based analysis, this study employs an Autoregressive Integrated Moving Average (ARIMA) forecasting model to evaluate the persistence and short-run predictability of UK inflation over time.

Forecasting inflation remains highly important for both policymakers and financial markets, as accurate inflation expectations play a central role in monetary policy decisions, wage negotiations, interest rate setting, and broader macroeconomic planning. Given the heightened volatility observed within the UK economy following the COVID-19 pandemic and global energy price shock, forecasting techniques provide additional insight into the extent to which historical inflation dynamics can explain future inflation behaviour.

The ARIMA framework was selected due to its widespread use within macroeconomic forecasting and time-series analysis. ARIMA models are specifically designed to capture autocorrelation, persistence, and dynamic behaviour within economic data by modelling relationships between current observations and their historical values. In the context of inflation analysis, ARIMA models are particularly useful because inflation often exhibits strong persistence effects, whereby previous inflation rates influence future inflation outcomes over time.

To estimate the forecasting model, the monthly CPI inflation series was transformed into a time-series object within RStudio and divided into separate training and testing samples. Observations covering the period from 2015 to 2023 were used to estimate the forecasting model, while inflation observations from 2024 were reserved for out-of-sample forecast evaluation. This approach allows the forecasting performance of the ARIMA model to be assessed by comparing predicted inflation values against realised inflation outcomes.

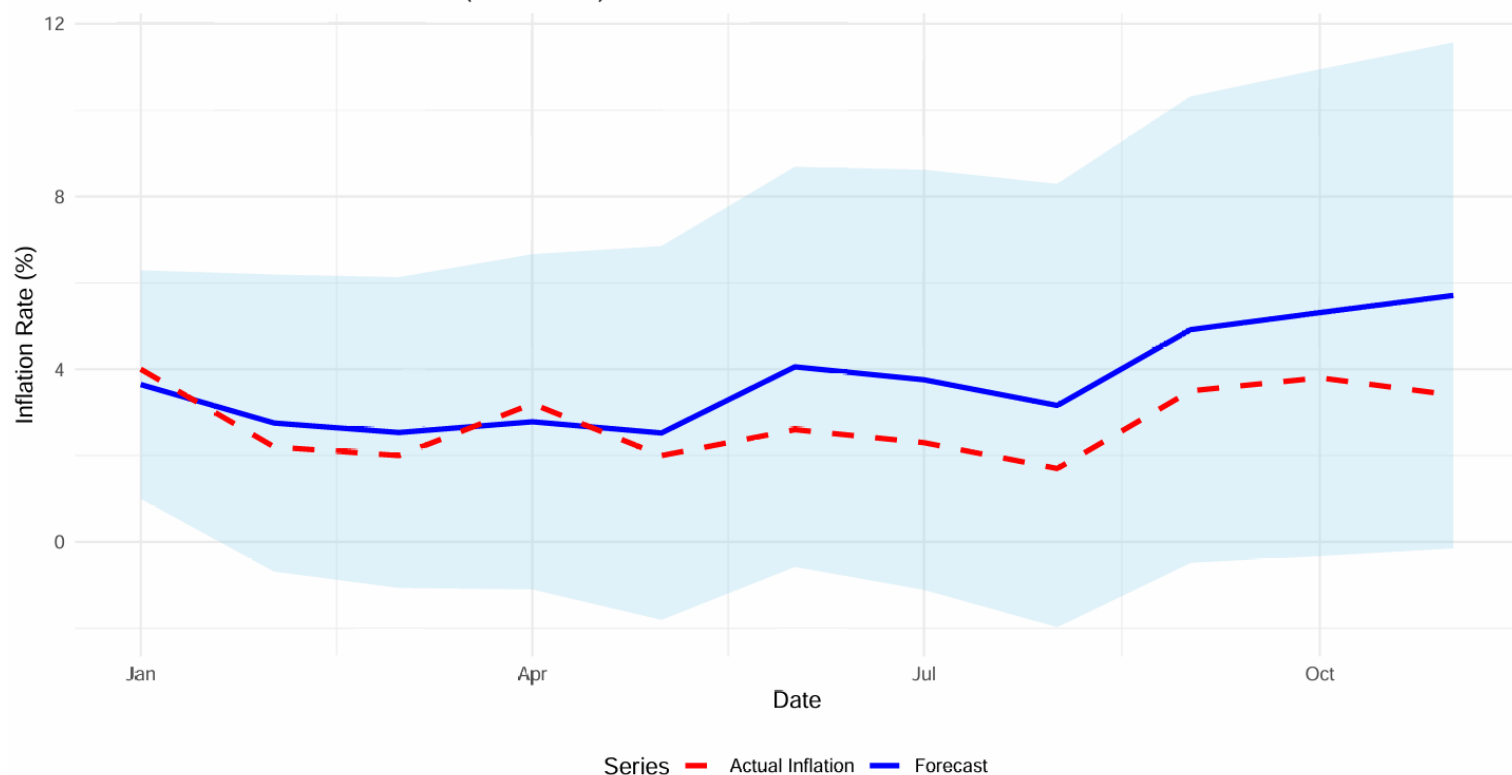
The optimal ARIMA specification selected by the model was ARIMA (2,1,0) (0,0,1) [12], indicating the presence of autoregressive dynamics, first differencing procedures to address non-stationarity, and seasonal moving average effects within the monthly inflation series. The inclusion of differencing is particularly important given the earlier diagnostic evidence suggesting that inflation exhibited non-stationary behaviour in levels. Consequently, the ARIMA model provides a more appropriate framework for capturing inflation persistence and dynamic adjustment processes over the sample period.

The following **Figure 1** presents the ARIMA forecast of UK CPI inflation alongside realised inflation outcomes and associated confidence intervals for the forecast horizon.

Figure 1:

UK Inflation Forecast Using ARIMA Model

Forecasted vs Actual CPI Inflation (2024–2025)



Source: ONS CPI Data and Author's ARIMA Forecast

Figure 1 presents the ARIMA forecast of UK CPI inflation alongside realised inflation outcomes over the 2024 forecast horizon. Overall, the forecasting model captures the broad direction and persistence of UK inflation reasonably effectively, particularly during the earlier stages of the forecast period. The forecasted inflation path generally follows the underlying movement of realised inflation, suggesting that the ARIMA framework successfully captures important persistence effects and dynamic inflation behaviour embedded within the historical CPI series.

However, the figure also highlights several important limitations associated with inflation forecasting during periods characterised by heightened macroeconomic volatility and structural uncertainty. While the ARIMA model captures the persistence of inflationary pressures, forecasted inflation begins to diverge from realised inflation outcomes during the latter stages of the forecast horizon. In particular, the model overpredicts inflation persistence toward the end of the sample period, projecting a renewed upward movement in inflation while realised inflation remains comparatively lower and more stable.

One explanation for this divergence is that statistical forecasting models such as ARIMA rely heavily upon historical inflation behaviour and persistence patterns when generating future forecasts. Consequently, unexpected structural changes, external policy interventions, or shifts in economic expectations may reduce forecasting accuracy during highly volatile macroeconomic environments. The post-pandemic inflation period was characterised by substantial uncertainty surrounding global energy markets, supply-chain disruption, geopolitical instability, and rapidly changing monetary policy conditions, all of which remain inherently difficult for univariate time-series models to fully capture.

The widening confidence intervals observed throughout the forecast horizon further reinforce the increasing uncertainty associated with inflation forecasting over time. As the forecast extends further into the future, the range of potential inflation outcomes becomes increasingly broad, reflecting the unpredictable nature of inflation dynamics during periods of elevated volatility and external economic shocks. This is particularly relevant within the context of recent geopolitical tensions and energy market instability, where disruptions to global oil supply chains and broader commodity markets may rapidly alter inflation trajectories beyond the scope of historically estimated statistical relationships.

An additional interpretation of the forecasting results relates to the role of expectations and behavioural feedback effects within the inflation process. The tendency of the ARIMA model to overpredict inflation persistence during the later stages of the forecast horizon may itself reflect the heightened uncertainty embedded within post-pandemic inflation dynamics. In periods where elevated inflation is expected to persist, household confidence and consumer sentiment may deteriorate further, contributing to weaker economic expectations and heightened macroeconomic instability.

This creates the potential for a behavioural inflation feedback mechanism whereby expectations of persistently high inflation negatively affect consumer confidence, while declining confidence simultaneously reinforces inflationary instability through shifts in spending behaviour, wage demands, precautionary saving, and expectations formation. In this context, the strong statistical significance of consumer confidence within both the interaction and dynamic inflation models becomes particularly important, suggesting that household sentiment played an increasingly influential role within post-pandemic UK inflation dynamics.

Consequently, the forecasting results suggest that inflation persistence following the pandemic was not driven solely by external supply-side shocks or labour market pressures, but also by evolving behavioural responses and expectations within the wider economy. The findings therefore reinforce the increasingly complex environment faced by central banks attempting to stabilise inflation during periods characterised by elevated uncertainty, geopolitical instability, and rapidly shifting macroeconomic expectations.

Overall, the forecasting analysis highlights both the usefulness and limitations of econometric forecasting models within modern macroeconomic environments. While ARIMA models remain valuable tools for identifying inflation trends and persistence dynamics, the findings also demonstrate that forecasting accuracy may deteriorate during periods characterised by structural breaks, external shocks, and rapidly evolving economic conditions. Consequently, the results reinforce the broader argument that modern inflation dynamics are increasingly shaped by complex interactions between domestic macroeconomic conditions, external supply-side disturbances, behavioural responses, and shifting inflation expectations.

5. Critical Evaluation

This study examined the determinants, persistence, and forecasting dynamics of UK inflation between 2015 and 2025 using a combination of econometric regression analysis and time-series forecasting techniques within

RStudio. By incorporating labour market variables, energy prices, and consumer confidence into both static and dynamic econometric frameworks, the analysis sought to evaluate how inflation behaviour evolved during a period characterised by substantial macroeconomic disruption, including Brexit-related uncertainty, the COVID-19 pandemic, the global energy price shock, and the subsequent inflation surge experienced across advanced economies.

The empirical findings suggest that UK inflation over the sample period was driven by a complex interaction between traditional labour market pressures, external supply-side shocks, and evolving behavioural responses within the wider economy. The baseline regression results provided strong evidence supporting conventional Phillips Curve relationships, with wage growth exerting a positive and statistically significant effect on inflation while unemployment maintained a negative relationship consistent with labour market inflationary pressures. Energy prices also demonstrated a significant positive relationship with inflation, reinforcing the importance of global commodity markets and external cost-push inflation during the post-pandemic recovery period.

However, one of the most important findings of the study is that inflation dynamics changed substantially following the COVID-19 pandemic. The interaction model demonstrated that wage growth and labour market conditions became significantly more influential within the inflation process after 2020, suggesting that inflationary pressures became increasingly embedded within domestic economic behaviour rather than remaining purely externally driven. The results therefore provide evidence of a steepening Phillips Curve relationship during periods of elevated inflationary pressure, supporting more recent macroeconomic literature arguing that inflation-output trade-offs intensify when inflation expectations become less anchored and supply-side shocks propagate more persistently throughout the economy.

The significance of consumer confidence across both the interaction and dynamic inflation models represents another particularly important contribution to the analysis. While traditional inflation models often focus primarily on labour markets and commodity prices, the findings of this study suggest that household sentiment and behavioural expectations played an increasingly important role within post-pandemic UK inflation dynamics. The deterioration in consumer confidence observed during periods of elevated inflation may itself have reinforced macroeconomic instability through expectations formation, precautionary behaviour, and shifts in wage and spending decisions. Consequently, the results indicate that inflation persistence may not solely reflect mechanical price adjustment processes, but also broader behavioural and psychological responses within the economy.

The dynamic inflation model further reinforced this conclusion by identifying strong evidence of inflation persistence through the significance of lagged CPI inflation. This finding suggests that inflationary shocks within the UK economy became partially self-reinforcing over time, increasing the difficulty faced by policymakers attempting to restore price stability once inflation expectations were adjusted upward. In this context, the forecasting analysis demonstrated both the strengths and limitations of econometric forecasting models during volatile macroeconomic periods. Although the ARIMA framework successfully captured broad inflation persistence and trend behaviour, the tendency to overpredict inflation during later forecast periods highlighted the challenges associated with modelling inflation in environments characterised by structural breaks, geopolitical uncertainty, and rapidly evolving economic conditions.

From a policy perspective, the findings highlight the increasingly complex environment confronting central banks within modern inflationary episodes. The interaction between external supply shocks, labour market tightness, inflation persistence, and deteriorating consumer confidence creates significant challenges for

monetary authorities attempting to balance inflation stabilisation alongside output and employment objectives. Geopolitical tensions surrounding global energy markets, including concerns relating to oil supply disruption and instability across strategically important trade routes, further reinforce the vulnerability of modern inflation dynamics to prolonged external shocks. Under such conditions, aggressive monetary tightening may reduce inflationary pressures while simultaneously weakening economic growth and household confidence, thereby intensifying stagflationary risks within the wider economy.

Nevertheless, several limitations should be acknowledged. While the econometric models employed within this study provide valuable insight into UK inflation dynamics, the analysis remains constrained by the availability and consistency of macroeconomic data across the selected sample period. Furthermore, the forecasting framework relies primarily upon historical statistical relationships and therefore may struggle to fully capture unexpected structural changes or exogenous geopolitical shocks. Future research may therefore benefit from incorporating nonlinear econometric models, expectations-based frameworks, or machine learning forecasting approaches capable of adapting more effectively to rapidly changing macroeconomic environments.

Overall, the findings of this study suggest that modern UK inflation dynamics cannot be explained solely through traditional macroeconomic relationships or external commodity shocks in isolation. Instead, inflation behaviour increasingly reflects the interaction between labour market conditions, supply-side disruption, persistence effects, and evolving behavioural expectations within the wider economy. Consequently, the analysis highlights the growing importance of incorporating both structural and behavioural considerations into contemporary inflation modelling and monetary policy evaluation during periods of heightened global uncertainty.

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Programming Language

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